Waterfall Model

The waterfall model is the ~~eas~~ earliest and one of the simplest software development methodology.

characteristis:

→ sequential development

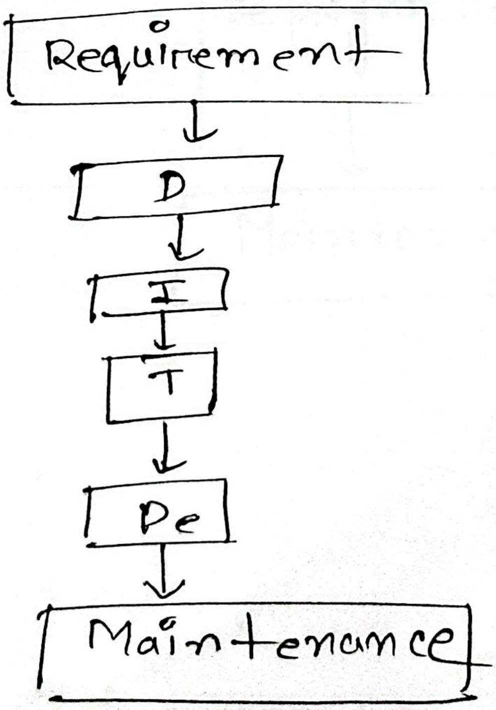
→ one phase is completed before the next begins.

→ Heavy documentation.

→ suitable for stable requirements.

→ Each phase flows downdward like a waterfall.

Process:



The output of one phase becomes the input to the next phase.

[4] Review meeting are conducted before moving to the next phase to detect errors early. Although waterfall is sequential, limited feedback between adjacent phases may occure to correct errors.

Limitation / Disadvantage:

(i) Difficult to accomodate changing requirements
(ii) Customer involvement is limited.
(iii) Testing occures late.
(iv) Errors may be expensive to fix.
(v) working software is delivered only at the end.

If customer requirments change during development, returning to earlier phases becomes costly.

(vi) Long development cycle before delivery.

* When should use waterfall?

→ Requirements are clear and stable.

→ Technology is well understood.

→ Project scope is fixed.

→ strong documentation is required.

→ Customer requirements are unlikely to change.

"Agile Development"

Agile is a modern development methodology that emphasizes flexibility and customer collaboration. It delivers software in small increments.

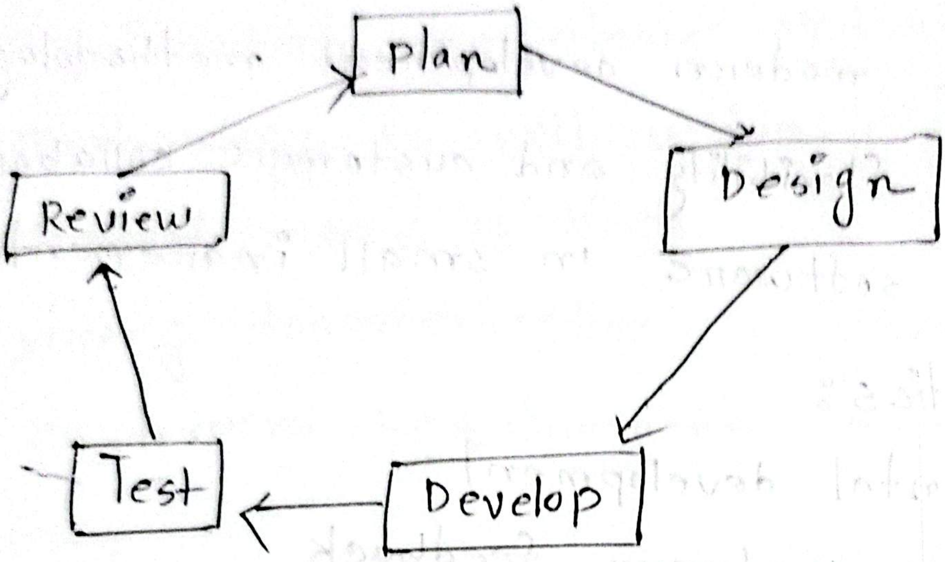
characteristics:

- Incremental development
- Continuous customer feedback
- Frequent software releases.
- Adaption to changing requirements.

Agile development encourages:

- Customer satisfaction.
- Small and frequent releases.
- Team collaboration
- Simplicity.
- continous improvement
- Flexibility.
- Rapid feedback.
- High-quality software.

cycle:



~~Eea~~ each iteration produces a working version of the software.

Extreme Programming (XP)

XP is one of the most popular Agile software development methodologies.

It was introduced by Kent Beck in 1999.

XP focuses on:

- Customer satisfaction
- Frequent software releases
- High-quality code
- Team collaboration
- Continuous improvement.

12 practices of XP:

① Planning Game:

- Requirements are written as User stories.
- Customers prioritize the features.
- Developers estimate effort and time.
- Team decides what will be developed next.

(ii) Small releases:

- Deliver software frequently.
- Each ~~relase~~ release provides business value
- Customer feedback is obtain early.

(iii) Metaphor:

- Uses a common story or analogy.
- Helps everyone understand the system
- Improve communication among the team member.

(iv) simple design:

(i) Design only what is needed now.
(iv) Avoid unnecessary complexcity.
• Improve the design later through refactoring.

(v) TDD (Test - Driven Development):

In xp testing begins before coding.

Benefits:

- Better software quality
- Early bug detection
- Easier maintenance.

(vi) Refactoring:

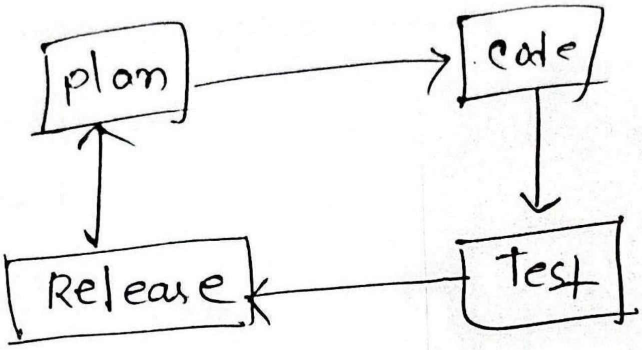
improve existing code without changing it's behavior.

(vii) Pair programming :

Two programmer work together at one computer.

- Driver writes the code.
- Navigator reviews the code continuously.

XP work



When use XP :

(1) Requirement changes frequently

(2) customer feedback

* (3) Project size small or medium

* (4) Developer works closely together

(v) Fast delivery is important.

core five values of XP:

1. communication
2. simplicity
3. Feedback
4. Courage
5. Respect.